Space complexity

Exercise sheet 11 (solutions)

Exercise 11.1. Prove that if $A, B \in \mathrm{L}$, then $A \cup B \in \mathrm{L}$ and $A \cap B \in \mathrm{L}$.

Solution. Run the two deterministic logspace deciders one after the other, reusing the same work tape. Accept if either accepts to decide $A \cup B$; accept if both accept to decide $A \cap B$. The combined machine still uses logarithmic space. □

Exercise 11.2. Consider the alphabet $\{(,)\}$ of parentheses. A word is balanced if every prefix has at least as many left parentheses as right parentheses, and the whole word has equal numbers of each. Prove that the language of balanced-parentheses words is in L.

Solution. Scan the input from left to right, keeping a counter d for the number of left parentheses minus the number of right parentheses seen so far. Increase it on '(', and decrease it on ')'. Reject if it ever becomes negative, and accept at the end exactly when it is zero. The counter is between 0 and the input length, so it uses $O(\log n)$ bits. This checks both the prefix condition and the equality of the total counts. □

Exercise 11.3. Let A, B, and C be languages. Prove the following statements about logspace reductions:

(a) If $A \leq_L B$ and $B \leq_L C$, then $A \leq_L C$.

(b) If $A \leq_L B$ and $B \in \mathrm{L}$, then $A \in \mathrm{L}$.

Solution. For (a), let r_1 reduce A to B and r_2 reduce B to C. Their composition preserves membership. To compute it in logarithmic space, simulate the transducer for r_2 on the virtual input $r_1(x)$. Whenever it asks for the symbol at a particular input position, rerun r_1 on x and count its output symbols until that position. The output of r_1 has polynomial length, so the position counter uses $O(\log |x|)$ bits. The simulation therefore computes a logspace reduction from A to C.

For (b), run the logspace decider for B on the virtual input $r(x)$ in the same way, producing symbols of $r(x)$ on demand. It accepts exactly when $r(x) \in B$, which holds exactly when $x \in A$. Thus $A \in \mathrm{L}$. □

Exercise 11.4. * Prove that $L \in \text{NP}$ if and only if a polynomial-time nondeterministic Turing machine decides L.

Solution. Suppose $L \in \text{NP}$, with verifier V and a polynomial bound $p(n)$ on certificate length. A nondeterministic machine guesses a certificate of length at most $p(n)$ and runs V. Every branch runs in polynomial time, and an accepting branch exists exactly when some certificate makes V accept.

Conversely, suppose a nondeterministic machine M decides L and halts within $q(n)$ steps on every branch, for a polynomial q. Since M is fixed, there is a constant bound on the number of possible transitions at each step. A certificate can record the choices along a branch using $O(q(n))$ bits. A deterministic verifier simulates M following those choices and accepts exactly when the branch accepts. Hence certificates have polynomial length and can be checked in polynomial time, so $L \in \text{NP}$. □

Exercise 11.5. * Let NEXP be the class of languages decided by exponential-time nondeterministic Turing machines. Prove that if P = NP, then EXP = NEXP.

Solution. Clearly $\text{EXP} \subseteq \text{NEXP}$, since a deterministic machine is a nondeterministic machine with only one available move. For the reverse inclusion, define

$$B = \{\langle M, w, 1^t\rangle \mid M \text{ has an accepting computation on } w \text{ of at most } t \text{ steps}\}.$$

The unary time bound means that a computation of M can be described by a polynomial-length list of transition choices. A deterministic verifier can simulate those choices for at most t steps, so $B \in \text{NP}$. Under the assumption P = NP, the language B has a deterministic polynomial-time decider.

Let $L \in \text{NEXP}$, decided by a fixed nondeterministic machine M in at most $T = 2^{p(n)}$ steps on inputs of length n. On input w, construct $\langle M, w, 1^T\rangle$ and run the decider for B. This input has exponential length in n, so constructing it and running a polynomial-time algorithm on it takes exponential time in n. The answer is yes exactly when M has an accepting branch on w. Thus $L \in \text{EXP}$, proving $\text{NEXP} \subseteq \text{EXP}$ and hence equality. □

Exercise 11.6. Let NL be the class of languages decidable by a nondeterministic Turing machine in logarithmic space. Prove that the following problem DIRECTED-REACHABILITY belongs to NL.

Input: Directed graph G, nodes s and t.

Question: Is there a path from s to t?

Solution. Store the current vertex, initially s, and a counter. At each step, nondeterministically guess an outgoing edge and move to its endpoint; accept if the current vertex is t. Stop after at most $|V| - 1$ steps. If a path exists, one with no repeated vertices exists and has at most $|V| - 1$ edges. The current vertex and counter use $O(\log |V|)$ bits, and checking a guessed edge takes logarithmic space. Thus this is an NL decider. □

Exercise 11.7. * Prove that the following problem NON-BIPARTITE belongs to NL.

Input: Graph G.

Question: Is G *not* bipartite?

Solution. An undirected graph is not bipartite exactly when it contains an odd cycle. Nondeterministically guess a starting vertex, then guess a walk that returns to it after an odd number of steps, checking each edge as it goes. Stop after at most $|V|$ steps. Store the start and current vertices, a counter, and the parity of the walk length; these use logarithmic space. An odd cycle gives an accepting branch. Conversely, every odd closed walk contains an odd cycle, so the machine accepts only non-bipartite graphs. Therefore NON-BIPARTITE is in NL. □

Exercise 11.8. * Prove that $\mathrm{P} \neq \mathrm{SPACE}(n)$.

Solution. By the space hierarchy theorem, choose a language $B \in \mathrm{SPACE}(n^2) \setminus \mathrm{SPACE}(n)$. Pad its inputs and define

$$B' = \{x\#0^{|x|^2} \mid x \in B\}.$$

To decide B', check the padding and simulate the decider for B on x. This uses $O(|x|^2)$ space, which is linear in the padded input length. Thus $B' \in \mathrm{SPACE}(n)$.

If $\mathrm{P} = \mathrm{SPACE}(n)$, then $B' \in \mathrm{P}$. Given x, construct $x\#0^{|x|^2}$ and run the polynomial-time decider for B'. The padded input has polynomial length in $|x|$, so this decides B in polynomial time. Hence $B \in \mathrm{P} = \mathrm{SPACE}(n)$, a contradiction. Therefore $\mathrm{P} \neq \mathrm{SPACE}(n)$. □